

Please upload for every problem a single pdf file named **Group_Pr_X_Last Name_First name{s}_CS_en_3Feb2021.pdf** where Group is the number of your group (ex 30311) and X is the number of problem. Please upload **only** in the appropriate assignment.

To be considered the problem, the **turn in** option should be checked.

Linear algebra, winter 2021, CS.

Problema 2

1. Find the dimension and a basis of the sum $S + V$ and intersection $S \cap V$ of the linear subspaces S and V if:

$$S = \text{span}\{(1, -1, 0, 0), (1, 1, 2, -2), (0, 1, 1, -1), (2, 2, 3, -3)\},$$

and

$$V = \text{span}\{(1, 0, 1, 0), (2, -2, 1, 0), (0, 1, 1, 1)\}.$$

2. Find a basis in $(S \cap V)^\perp$.

Note. In the file submitted please write computations, explanations and everything needed for a better understanding of the problem.